

## 1.2 Glaciated Landscapes

Geography — Eduqas A-Level (A110QS) — Component 1: Changing Landscapes

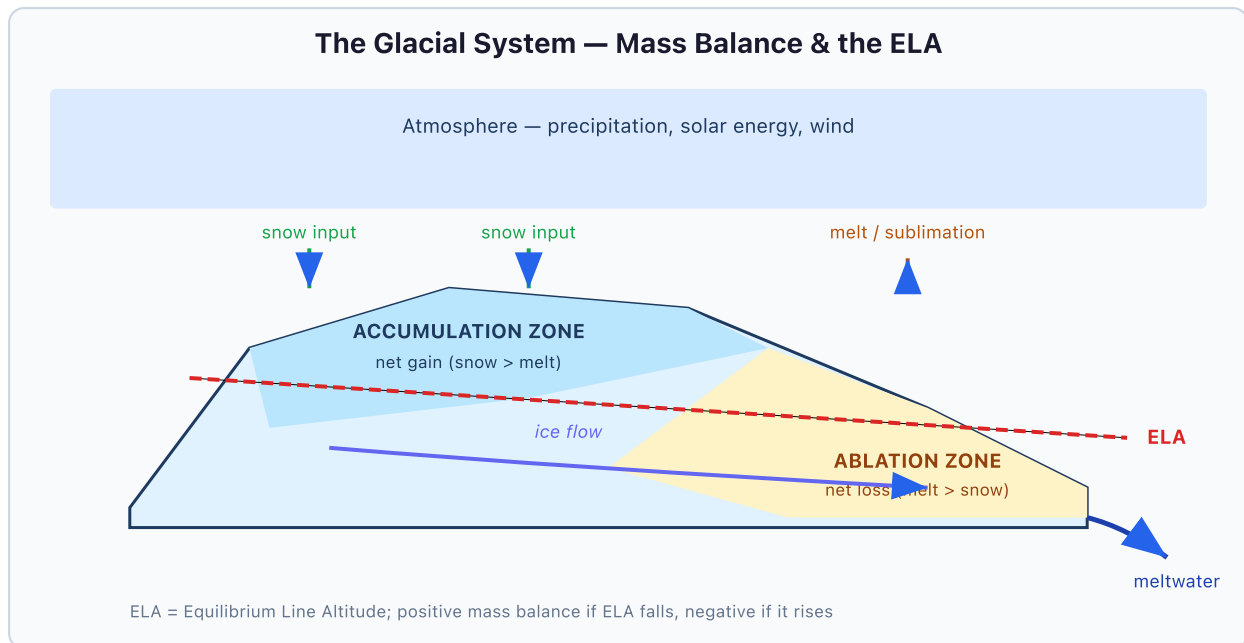
### 1. The Glacial System

Eduqas requires glaciated landscapes to be approached as an **open system** in which mass (snow, ice, water, sediment) and energy (solar radiation, gravitational potential, latent heat) are exchanged with the atmosphere, lithosphere and hydrosphere. The system tends toward a *dynamic equilibrium* measured by **mass balance**: the net gain or loss of ice over an annual cycle. A glacier is not merely a body of ice — it is a conveyor belt that receives snow at its head, transforms it through firn to glacier ice, flows downslope under gravity, and releases meltwater and sediment at its snout. Landforms record the interaction of this flow with the underlying bedrock and with sediment released from the ice.

**Glacial system:** An open system comprising inputs (precipitation as snow, avalanche, aeolian dust), stores (snowfield, firn, ice), flows (accumulation, ablation, ice movement) and outputs (meltwater, icebergs, sediment). The system seeks dynamic equilibrium between accumulation and ablation.

#### System components:

- **Inputs** — precipitation (snow), avalanches from surrounding slopes, rockfall debris, aeolian dust deposited on the ice surface, and condensation/refreezing of meltwater.
- **Stores** — the snowfield (fresh snow, density  $\sim 100 \text{ kg/m}^3$ ), firn/névé (partially compacted granular snow,  $400\text{--}830 \text{ kg/m}^3$ ), and glacier ice (density  $\sim 900 \text{ kg/m}^3$ ).
- **Transfers/flows** — accumulation (snowfall + densification), ablation (melting, sublimation, calving), basal sliding, internal deformation, englacial and subglacial meltwater flow.
- **Outputs** — meltwater (pro-glacial streams), icebergs from calving fronts, glacial till and fluvio-glacial sediment, long-wave radiation and latent heat.



**Glacier mass balance:** The net annual difference between accumulation (inputs of snow and ice) and ablation (outputs by melting, sublimation and calving) for the whole glacier. A *positive* mass balance year causes the glacier to advance or thicken; a *negative* mass balance year causes retreat or thinning. Most glaciers worldwide have been in persistent negative mass balance since the 1980s.

**Equilibrium Line Altitude (ELA):** The altitude at which annual accumulation exactly equals annual ablation, separating the upper accumulation zone from the lower ablation zone. A rising ELA (climate warming) shrinks the accumulation area and produces retreat; a falling ELA produces advance. The ratio of accumulation area to total area (AAR) is typically 0.6–0.7 for a glacier in equilibrium.

**Exam tip — always open with the system:** Whether the command word is *explain*, *assess* or *to what extent*, begin essays by defining the glacier as an open system in dynamic equilibrium and reference the ELA. Examiner reports consistently flag candidates who jump straight into landforms without the systems framework as losing AO1 marks.

**Exam tip — mass balance = annual ledger:** Always describe mass balance as an annual accounting exercise: accumulation (credits) vs ablation (debits). A glacier with three successive positive-balance years will thicken and advance after a lag of a few to several decades (the

response time). This lag explains why glaciers retreating today reflect warming of the 1950s to 1980s, not just the last decade.

## Snow to Ice Transformation

The transformation of freshly fallen snow to glacier ice is progressive and takes decades in temperate settings, centuries in cold polar settings. Fresh snow has a density of  $\sim 50\text{--}200\text{ kg m}^{-3}$  and consists of delicate hexagonal crystals with abundant air space. Wind-packing, diurnal melt-refreeze, pressure from overlying snow and kinetic grain-boundary sublimation progressively destroy these crystal forms, producing a granular mass called *firn* or *névé* (density  $400\text{--}830\text{ kg m}^{-3}$ ). Continued burial collapses the remaining air-space until the material reaches the *pore close-off* density of  $\sim 830\text{ kg m}^{-3}$ , at which point it becomes true glacier ice — an impermeable, plastic crystalline solid containing only isolated air bubbles. These trapped bubbles are the primary archive used in ice-core palaeoclimate reconstruction (Vostok, EPICA Dome C, GRIP, NEEM).

Snow ( $100\text{ kg m}^{-3}$ )  $\rightarrow$  Firn ( $550\text{ kg m}^{-3}$ )  $\rightarrow$  Glacier ice ( $900\text{ kg m}^{-3}$ )

The **firn line** on a glacier surface, visible at the end of the summer melt season, approximates the ELA: above it, snow from the previous winter survives to become next year's firn; below it, bare glacier ice is exposed. Candidates should note that the ELA, AAR (accumulation area ratio) and firn line are all quantitative proxies for long-term mass balance.

## 2. Classification of Ice Masses

Eduqas requires familiarity with two orthogonal classifications: by **form** (size and topographic setting) and by **thermal regime** (temperature of the basal ice). The thermal regime is the more important control on process, because it determines whether basal sliding — and therefore meaningful erosion — can occur.

### Classification by Form

| Type | Description | Example |
|------|-------------|---------|
|------|-------------|---------|

|                       |                                                                   |                                                                                    |
|-----------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Continental ice sheet | >50,000 km <sup>2</sup> ; covers whole continents; flows radially | Antarctica (14 million km <sup>2</sup> ), Greenland (1.7 million km <sup>2</sup> ) |
| Ice cap               | <50,000 km <sup>2</sup> ; dome-shaped; buries underlying relief   | Vatnajökull (Iceland, ~7,900 km <sup>2</sup> )                                     |
| Ice field             | Non-domed, controlled by relief; high peaks protrude (nunataks)   | Southern Patagonian Icefield                                                       |
| Valley glacier        | Flows between valley walls; fed by cirques or icefield            | Aletsch (Switzerland), Mer de Glace (France)                                       |
| Cirque glacier        | Small, confined to armchair hollow on mountainside                | Cwm Idwal (historic), many in Rockies                                              |
| Piedmont glacier      | Valley glacier spills onto lowland and spreads into lobe          | Malaspina Glacier, Alaska                                                          |
| Ice shelf             | Floating extension of ice sheet over sea                          | Ross Ice Shelf, Larsen C (Antarctica)                                              |

## Classification by Thermal Regime

| Regime                  | Basal temperature                           | Basal sliding?                 | Erosion          | Examples                               |
|-------------------------|---------------------------------------------|--------------------------------|------------------|----------------------------------------|
| Temperate (warm-based)  | At pressure melting point (PMP)             | Yes — dominant movement        | Highly effective | Alps, Southern Alps NZ, Patagonia      |
| Polar (cold-based)      | Below PMP; frozen to bed                    | No — only internal deformation | Very limited     | Central Antarctica, interior Greenland |
| Sub-polar / polythermal | Frozen at margins, melting beneath interior | Partial                        | Intermediate     | Svalbard, Arctic Canada                |

**Basal sliding:** Movement of a glacier over its bed, lubricated by a thin film of meltwater produced when pressure lowers the melting point of ice (pressure melting) at the bed. Basal sliding only

occurs in warm-based (temperate) glaciers and accounts for up to 75% of their total movement. Cold-based glaciers are frozen to the bed and move only by internal deformation, producing minimal erosion.

**Exam tip — thermal regime drives erosion:** When asked to explain why the Alps show more dramatic cirques and troughs than central Antarctica, the key point is thermal regime, not ice thickness. Temperate ice erodes; cold ice protects underlying relief. Many candidates mistakenly assume more ice equals more erosion.

**Exam tip — do not confuse “glacier” with “ice sheet”:** Examiner mark schemes penalise imprecise terminology. An *ice sheet* covers an entire continent and flows radially; an *ice cap* is smaller and dome-shaped; a *valley glacier* is confined between valley walls; a *cirque glacier* occupies an armchair hollow. Always use the precise term that fits the case study you are naming.

## Glaciations in the Quaternary

The Quaternary Period (2.58 Ma to present) is defined by repeated glacial-interglacial cycles driven by Milankovitch orbital forcing (eccentricity ~100 ka, obliquity ~41 ka, precession ~23 ka). At least 40 glacial cycles are recorded in marine  $\delta^{18}\text{O}$  records; the Last Glacial Maximum (LGM, ~21 ka BP) saw the British Ice Sheet reach as far south as north Norfolk and south Wales, North American Laurentide Ice Sheet to the modern Ohio River, and global sea level drop ~130 m. The Holocene (11.7 ka BP to present) is the current interglacial; the preceding *Younger Dryas (Loch Lomond Stadial)* ~12.8–11.7 ka BP produced the final limited re-advance of British glaciers and is responsible for the most visible relict landforms in Snowdonia, the Lake District and Western Scotland.

The concept of *glaciated* vs *glacial* is important: *glacial* means “currently under ice” (e.g. Antarctica, Iceland); *glaciated* means “formerly or presently shaped by ice” (e.g. Snowdonia, Norway). The Eduqas specification focuses on glaciated landscapes of both kinds and rewards accurate terminology.

## Ice-Mass Classification Summary Table

| Criterion | Warm-based (temperate) | Cold-based (polar) | Polythermal |
|-----------|------------------------|--------------------|-------------|
|-----------|------------------------|--------------------|-------------|

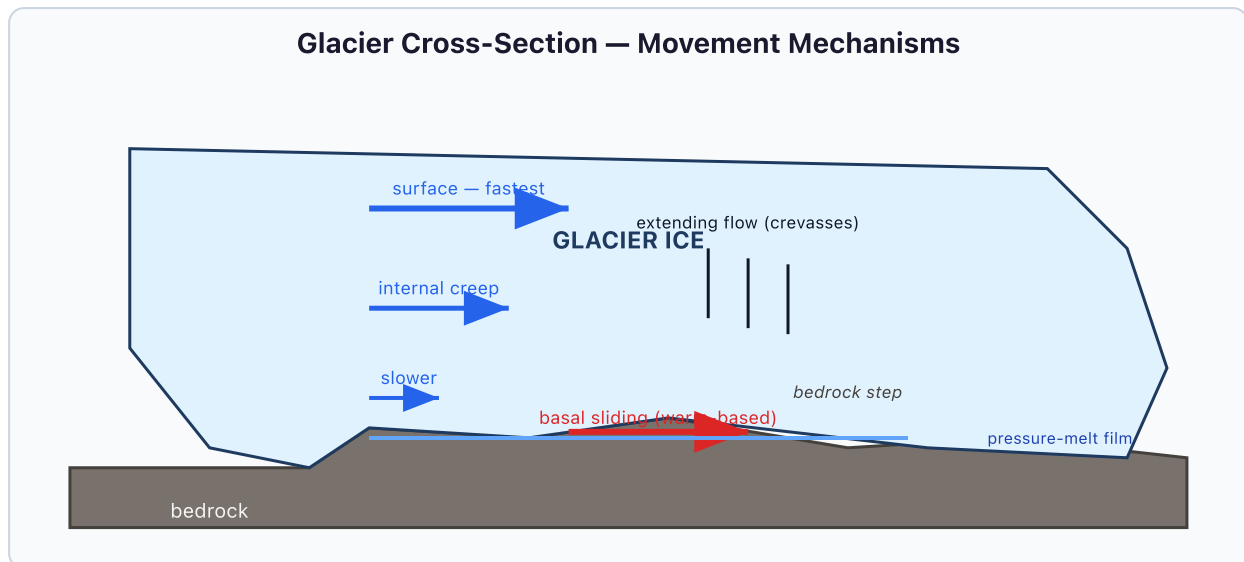
|                  |                                     |                                        |                         |
|------------------|-------------------------------------|----------------------------------------|-------------------------|
| Basal temp.      | At PMP ( $\sim 0^{\circ}\text{C}$ ) | < PMP                                  | Mixed                   |
| Meltwater at bed | Abundant                            | Absent                                 | Patchy                  |
| Basal sliding    | Yes, >50% of flow                   | None                                   | Partial                 |
| Erosion rate     | High ( $>1\text{ mm yr}^{-1}$ )     | Very low ( $<0.01\text{ mm yr}^{-1}$ ) | Intermediate            |
| Typical velocity | 10s–100s $\text{m yr}^{-1}$         | $<10\text{ m yr}^{-1}$                 | Variable                |
| Classic examples | Alps, Patagonia, Iceland            | Central E. Antarctica, inner Greenland | Svalbard, Arctic Canada |

### 3. Glacier Movement

Glacier ice is a crystalline solid that deforms plastically under its own weight. Movement is the product of two independent mechanisms:

1. **Internal deformation** — movement within the ice body by creep, intergranular slip and plastic flow. Dominant in cold-based glaciers and above the pressure melting point in all glaciers. Rates of  $\sim 1\text{--}20\text{ m yr}^{-1}$ .
2. **Basal sliding** — movement of the entire ice mass over its bed on a film of pressure-melt water; also *regelation slip* (localised refreezing downstream of obstacles). Only in warm-based ice. Rates of  $20\text{--}300\text{ m yr}^{-1}$ , higher in outlet glaciers.

Where a glacier thins and accelerates over a steepening bedrock step, flow is *extending* and the surface cracks into transverse crevasses. Where it thickens and decelerates against a reverse bedrock slope, flow is *compressive* and surface thrust faults and shear planes develop. Crevasse patterns therefore allow interpretation of bedrock geometry from the surface alone.



## Surging Glaciers

A **surge** is a temporary, dramatic acceleration (up to 100× normal velocity) caused by build-up of subglacial meltwater, which reduces basal friction. Meier and Post (1969) described a classic surge cycle: a long *quiescent phase* of slow flow and upstream thickening is followed by rapid *surge phase* lasting 1–7 years. Examples include Variegated Glacier (Alaska, surged 1982–83), Bruarjökull (Iceland, surges every ~80 years), and Kutiah (Pakistan). Surges are useful exam examples because they illustrate non-linear, threshold-driven glacier behaviour.

**Exam tip — velocity varies spatially and temporally:** Many students describe glacier flow as uniform. In reality, velocity is highest at the surface centre-line (least friction) and lowest at the bed and valley walls. Seasonally, summer meltwater injection accelerates flow, whereas winter freezing slows it. Reference these variations for top-band marks in 12/20-mark questions.

## Factors Influencing Velocity

- **Ice thickness** — greater weight increases basal shear stress, accelerating internal deformation (deformation rate varies as approximately the *third power* (cube) of the applied stress: Glen's Flow Law).
- **Bed slope** — steeper gradients increase the gravitational driving stress.
- **Bed temperature** — above PMP allows sliding; below PMP restricts movement to creep.
- **Basal meltwater availability** — lubricates the bed and reduces friction; seasonal meltwater pulses cause summer speed-ups of 20–200%.



- **Bed roughness** — smooth beds favour sliding; rough beds require internal deformation around obstacles.
- **Subglacial sediment deformation** — wet, fine-grained till can itself deform, acting as a layer of “bed lubrication” (e.g. Siple Coast Ice Streams, Antarctica).

$$\tau_b = \rho g h \sin(\alpha) \quad (\text{basal shear stress driving glacier flow})$$

where  $\rho$  = ice density ( $\sim 900 \text{ kg m}^{-3}$ ),  $g$  = gravity,  $h$  = ice thickness,  $\alpha$  = surface slope.

## 4. Glacial Processes

### Erosion

Two main mechanisms, both requiring warm-based ice:

- **Plucking (quarrying):** Meltwater percolates into joints in the bedrock, freezes onto the base of the ice, and is then pulled away as the glacier advances, extracting large clasts. Produces the jagged down-ice face of a *roche moutonnée* and the back wall of a cirque. Effective only where bedrock is well-jointed.
- **Abrasion:** Rock debris embedded in the basal ice acts as a rasp, scraping bedrock and producing fine rock flour, striations (parallel scratches) and polished surfaces. Effective only with warm-based ice that provides debris-loaded sliding contact.

Freeze-thaw weathering above the ice surface (on nunataks and cirque back walls) supplies angular debris onto the glacier via rockfall; this frost-shattered material becomes the abrasive tool of subsequent plucking and abrasion. Other processes include *dilation* (pressure release fracturing as ice unloads), *meltwater erosion* under the glacier and *nivation* on snow patches at the head of cirques.

**Plucking:** Glacial erosion by which meltwater freezes into joints in bedrock, bonding rock to basal ice; as the glacier moves downslope, the frozen-on rock is torn away. Produces the steep stoss-lee asymmetry of *roches moutonnées* and the angular back walls of cirques. Requires warm-based ice and well-jointed, often frost-shattered bedrock.



**Abrasion:** Glacial erosion by which rock debris held in basal ice scrapes the bedrock as the glacier slides over it, producing *striations* (mm-scale scratches), *grooves* (m-scale), polished surfaces and fine rock flour that turns meltwater streams milky. Most effective when abundant hard clasts are in contact with a softer substrate under moderate pressure.

## Factors Controlling Rate of Glacial Erosion

Five variables jointly determine how fast a glacier erodes:

1. **Presence of basal meltwater** — temperate ice erodes; cold-based ice protects.
2. **Ice velocity** — more passages of debris-armed ice per unit time means more abrasion.
3. **Debris concentration in basal ice** — few tools = little abrasion.
4. **Debris-clast hardness vs bedrock** — hard clasts on softer rock (e.g. quartzite on shale) erode efficiently.
5. **Bedrock jointing and weathering** — well-jointed, frost-weakened rock is readily plucked.

Glacial erosion rates (Hallet et al., 1996; Koppes et al., 2015) vary from  $<0.01 \text{ mm yr}^{-1}$  in cold-based parts of Antarctica to  $>30 \text{ mm yr}^{-1}$  in temperate glaciers of the Patagonian Andes — a four-orders-of-magnitude range entirely governed by the factors above.

## Transport

Debris is carried in three positions within the ice:

- **Supraglacial** — on the upper surface, derived from rockfall off valley walls. Visible as lateral and medial moraines.
- **Englacial** — within the ice body, either sinking supraglacial debris or material trapped as ice accumulates.
- **Subglacial** — at the ice-bed interface, material acquired by plucking and abrasion. This is the most geomorphically active zone.

## Deposition

Glacial deposition produces two distinct sediment types:

- **Till** (boulder clay) — unsorted, unstratified, angular sediment deposited directly by ice. Particle sizes range from clay to erratic boulders. Includes *lodgement till* (pressed into the bed beneath moving ice) and *ablation till* (released as ice melts).
- **Fluvio-glacial sediments** — sorted and stratified deposits laid down by meltwater. Reworked till, commonly rounded, arranged in beds by particle size. Found in eskers, kames, kame terraces and outwash plains.

**Exam tip — link process to landform every time:** Eduqas mark schemes explicitly credit process-to-landform sequences (plucking → cirque back wall; abrasion → U-shaped trough floor; meltwater → esker). A common error is to describe the landform only. Always name the process, say *how* it operates, then attribute it to the landform feature it creates.

## 5. Erosional Landforms

### Cirques (Corries, Cwms)

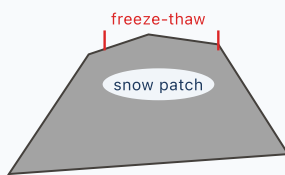
A cirque is an armchair-shaped hollow on a mountainside, marked by a steep (often over-steepened) back wall, a roughly flat or rotationally over-deepened floor, and a raised front lip that often dams a tarn after deglaciation. Formation is a sequence:

1. **Nivation** — a snow patch lying in a shaded hollow (usually north- or north-east-facing in UK) promotes freeze-thaw weathering around its margins, deepening the hollow (*nivation hollow*).
2. As snow thickens and compacts to firn and then ice, accumulation exceeds ablation in the hollow.
3. A small cirque glacier forms. Rotational slip beneath it deepens the hollow by abrasion, while plucking retreats the back wall.
4. The raised front lip survives because ice is thinnest and slowest there, so erosion is minimal — it may be partially constructional (moraine).
5. On deglaciation, the over-deepened basin fills with water to form a *tarn* (e.g. Llyn Idwal, Red Tarn).

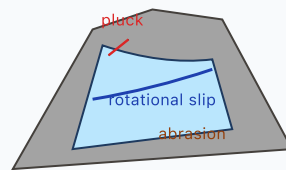
**Cirque (cwm/corrie):** An amphitheatre-shaped, glacially-eroded hollow on a mountain flank, with a steep plucked back wall, over-deepened basin floor produced by rotational abrasion, and a raised front lip. Many hold tarns after deglaciation. In the UK they are overwhelmingly north- to north-east-facing because that aspect preserves snow longest.

### Cirque Formation — Sequential Development

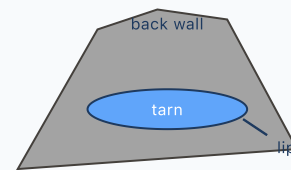
Stage 1: Nivation hollow



Stage 2: Active cirque glacier



Stage 3: Post-glacial tarn



Process: nivation → accumulation → rotational slip (abrasion + plucking) → over-deepened basin

UK example: Cwm Idwal, Snowdonia; Red Tarn, Helvellyn

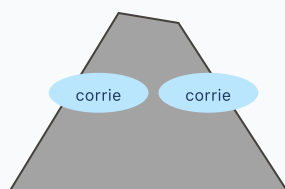
## Arêtes and Pyramidal Peaks

When two cirques erode back-to-back on either side of a ridge, the intervening ridge is narrowed to a **knife-edge arête** (e.g. Striding Edge, Helvellyn; Crib Goch, Snowdon). Where three or more cirques erode toward a common centre, the intervening ridge is reduced to an isolated **pyramidal peak (horn)** — the Matterhorn (4,478 m, Alps) is the archetype; Y Lliwedd and portions of Snowdon are partial examples in Britain.

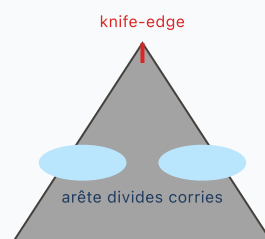
**Arête:** A sharp, knife-edged ridge of rock formed when two cirques erode back-to-back, progressively narrowing the intervening divide by headward plucking and freeze-thaw weathering above the ice.

### From Corrie to Arête to Pyramidal Peak

(a) Two corries enlarging



(b) Arête forms



(c) Pyramidal peak (horn)



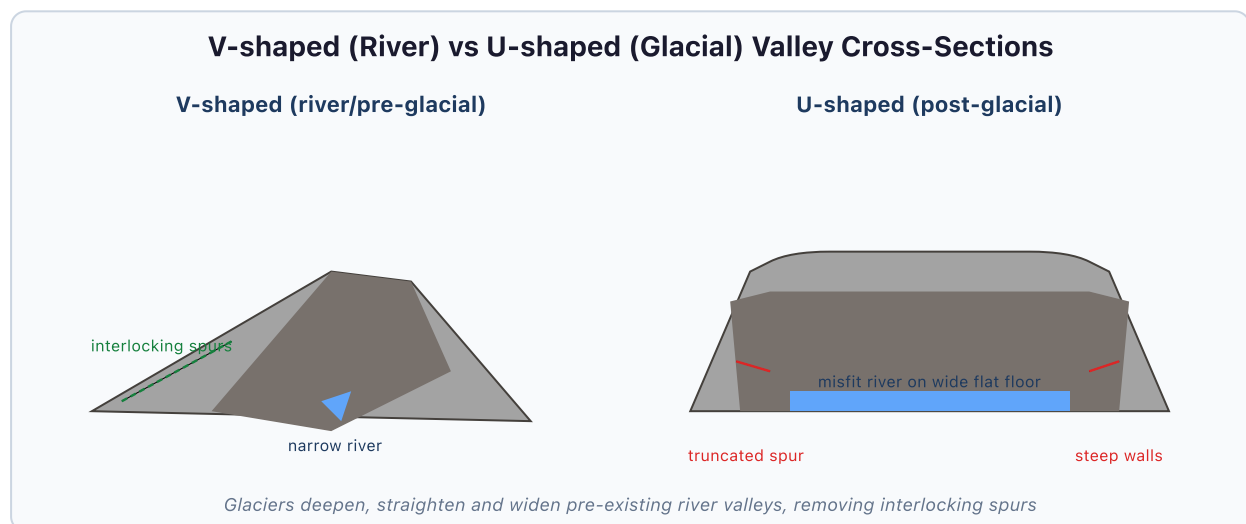
3+ cirques converging

Example: Striding Edge (arête) and Matterhorn (horn)

## Glacial Troughs (U-shaped Valleys)

Pre-glacial river valleys are profoundly modified by passing glacier ice. The valley floor is deepened by basal abrasion and plucking, the valley walls are steepened, interlocking spurs are removed as **truncated spurs**, and the cross-profile is reshaped from V-shaped to the characteristic **U-shape** (or, more accurately, *parabolic*). Long profiles show *over-deepened basins* (often occupied by ribbon lakes) and *rock steps* where more resistant rock remains.

**Glacial trough (U-shaped valley):** A deep, wide, steep-sided valley with a broad flat or rounded floor, formed when a glacier occupies and over-deepens a pre-existing V-shaped river valley. The cross-profile is characteristically U-shaped; the long profile shows over-deepened basins, rock steps and truncated spurs. Classic examples: Nant Ffrancon (Snowdonia), Lauterbrunnen (Alps), Yosemite (California).



## Formation of an Over-deepened Basin

Long-profile over-deepening of a glacial trough arises from three reinforcing controls: (i) rock resistance contrasts (softer rock erodes faster); (ii) flow convergence (where tributary ice joins the trunk); and (iii) selective basal meltwater generation (where thicker ice raises bed pressure, locally lowering the melting point and producing more sliding). The resulting sequence of rock steps (*Riegels*) and over-deepened basins is diagnostic of valley glaciation and commonly occupied by ribbon lakes when deglaciation is complete. In the Lake District, Windermere occupies such a basin and reaches a depth of 67 m, well below the surrounding valley floor.

## Hanging Valleys

A tributary glacier carries less ice than the trunk glacier and therefore erodes its valley floor to a shallower depth. After deglaciation the tributary valley is left perched above the main trough floor. Streams now descending from the tributary plunge as *waterfalls* over the step. UK examples include Cwm Idwal above Nant Ffrancon; global examples include Bridalveil Fall and many others in Yosemite. Hanging valleys are reliable indicators of former glaciation.

## Ribbon Lakes

Long, narrow lakes occupying glacial troughs, formed where the glacier selectively over-deepened a zone of weaker or more jointed rock, leaving an elongated basin. They may be dammed downstream by a terminal moraine. Classic UK examples: Wastwater, Windermere, Ullswater (Lake District); Llyn Padarn (Snowdonia). Ribbon lakes are frequently used in exam questions as an illustration of *differential erosion*.

## Roches Moutonnées

Small bedrock mounds up to tens of metres long, showing marked asymmetry: a smooth, gently sloping, often striated *up-ice* (*stoss*) side shaped by abrasion, and a steep, jagged, plucked *down-ice* (*lee*) side. Named from a supposed resemblance to wigs worn in 18th-century France (*moutonné*). Good examples: Central Park (New York) and many in Glen Coe. They record the direction of former ice flow.

## Striations and Chattermarks

Mm to cm-scale scratches aligned with ice-flow direction are *striations*; crescentic gouges produced by stick-slip motion of clasts are *chattermarks*. Together they supply a primary palaeo-flow indicator for mapping former ice directions in relict landscapes. In Snowdonia, striations on slate pavements near Capel Curig record a south-westward ice flow during the Younger Dryas.

## Crag and Tail

An elongated feature formed where resistant bedrock (the *crag*) protected softer rock or till (the *tail*) from glacial erosion on the down-ice side. Edinburgh Castle Rock (crag, volcanic plug) with the Royal Mile (tail, till) is the classic British example.

## Comparative Landform Table

| Landform | Formation process | UK example | Global example |
|----------|-------------------|------------|----------------|
|----------|-------------------|------------|----------------|

|                       |                                                       |                                    |                                 |
|-----------------------|-------------------------------------------------------|------------------------------------|---------------------------------|
| Cirque / cwm / corrie | Nivation → rotational slip → abrasion + plucking      | Cwm Idwal (Snowdonia)              | Western Cwm, Everest            |
| Arête                 | Back-to-back cirque back-wall retreat                 | Striding Edge (Helvellyn)          | Aiguille du Midi area, Chamonix |
| Pyramidal peak (horn) | ≥3 cirques converging                                 | Y Lliwedd / Snowdon                | Matterhorn (Alps)               |
| U-shaped trough       | Abrasion + plucking deepens and widens V-valley       | Nant Ffrancon (Snowdonia)          | Yosemite Valley (USA)           |
| Hanging valley        | Tributary glacier less erosive than trunk             | Cwm Idwal over Nant Ffrancon       | Bridalveil Fall, Yosemite       |
| Ribbon lake           | Over-deepened basin in trough                         | Wastwater (79 m deep)              | Lake Como, Lake Geneva          |
| Truncated spur        | Ice straightens valley, shears off interlocking spurs | Glen Coe                           | Lauterbrunnen Valley            |
| Roche moutonnée       | Abrasion up-ice, plucking down-ice                    | Central Scotland                   | Central Park, NYC               |
| Crag and tail         | Resistant “crag” protects softer lee                  | Edinburgh Castle Rock / Royal Mile | Stirling Castle                 |

## 6. Depositional Landforms

### Moraines

Moraines are accumulations of till forming ridges or mounds in specific positions relative to the glacier. Candidates are expected to recognise and explain:

- **Lateral moraine** — ridges of debris deposited along the glacier's sides, fed by frost-shattered rockfall from valley walls.

- **Medial moraine** — a ridge formed on the ice surface where two tributary glaciers converge and their lateral moraines merge.
- **Terminal (end) moraine** — a ridge at the furthest advance of the glacier, marking its maximum extent. Classic UK example: the Cromer Ridge, Norfolk (~30 m high, Anglian glaciation).
- **Recessional moraine** — a series of smaller ridges marking temporary still-stands during overall retreat.
- **Push moraine** — a ridge of previously deposited sediment bulldozed up by a short re-advance.
- **Ground moraine** — a blanket of unsorted till deposited beneath the glacier, forming undulating lowlands.

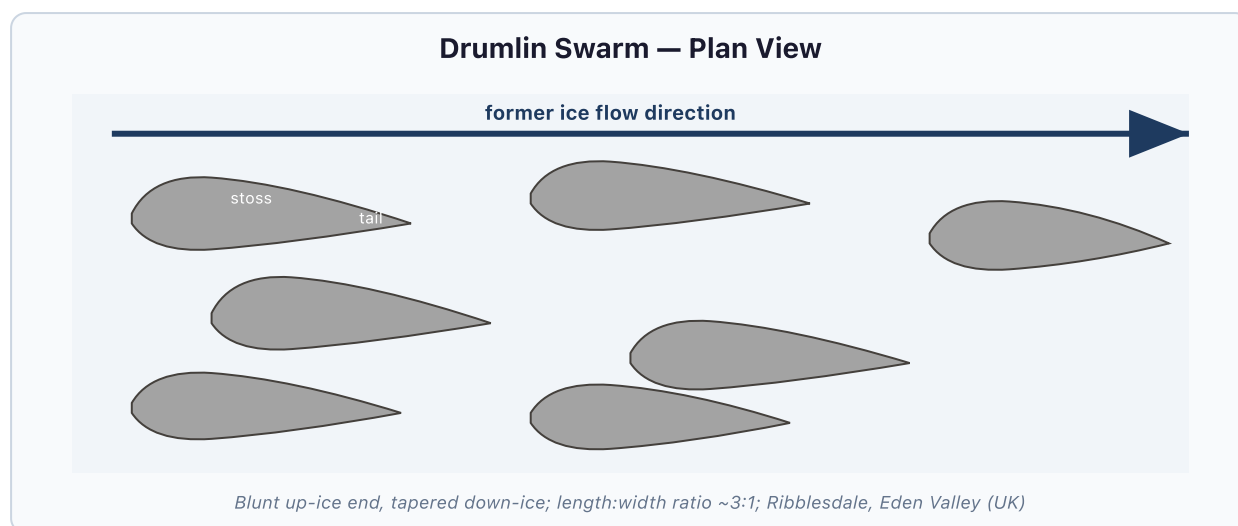
**Till (boulder clay):** Unsorted, unstratified sediment deposited directly by ice, containing a chaotic mix of particle sizes from clay and silt to angular boulders (erratics). Differs fundamentally from fluvio-glacial sediments, which are sorted and stratified by water. Till is the parent material of moraines and drumlins.

## Drumlins

Streamlined, elongated oval hills of till, with a blunt **stoss** end facing up-ice and a tapered **lee** tail pointing down-ice. Lengths 100–2,000 m; heights 5–50 m. Long axis to short axis ratios of 2.5:1 to 4:1. Typically form in *swarms* ("basket-of-eggs" topography) where shear stress at the glacier bed has remoulded till into the most hydrodynamically efficient shape for ice flow. Their long axes give reliable palaeo-flow directions.

**Drumlin:** A streamlined oval hill of glacial till, up to 2 km long and 50 m high, with the blunt end oriented up-ice and the tapered end down-ice. Formed beneath actively flowing ice, probably by remoulding of saturated subglacial till. Swarms occur in Ribblesdale, Eden Valley (Cumbria) and County Down (Northern Ireland).





## Summary: Erosional vs Depositional Landscapes

| Feature                      | Erosional landscape                   | Depositional landscape                    |
|------------------------------|---------------------------------------|-------------------------------------------|
| Relief                       | High, angular, scoured                | Subdued, rolling or flat                  |
| Dominant process             | Plucking + abrasion by warm-based ice | Melt-out, lodgement, meltwater sorting    |
| Sediment cover               | Thin, often bare bedrock              | Thick till, sand and gravel               |
| Typical setting              | Uplands close to accumulation zones   | Lowlands beyond maximum ice extent        |
| Classic diagnostic landforms | Cirques, arêtes, troughs              | Drumlins, moraines, eskers, outwash       |
| UK examples                  | Snowdonia, Lake District, Cairngorms  | East Anglia, Vale of York, Cheshire Plain |

## Erratics

Boulders of rock transported by ice and deposited far from their source outcrop. Because they can be lithologically traced, erratics are powerful indicators of past ice-flow patterns. Classic examples: the *Norber erratics* (Silurian sandstone boulders perched on Carboniferous limestone pedestals, Yorkshire

Dales), and the Norwegian erratics on the east coast of England, transported across the North Sea during the Devensian.

## Variants of Drumlins and Related Streamlined Forms

- **Classic till drumlin** — entirely till-cored, as at Ribblesdale.
- **Rock-cored drumlin** — bedrock knob streamlined by till accretion on its downstream tail.
- **Mega-scale glacial lineation (MSGSL)** — drumlin-like ridges >10 km long, found offshore on former ice-stream beds (e.g. Mælaren Bay, Baltic).
- **Crag and tail** — structural resistant rock as head; glacial till draped down-ice.
- **Fluted moraine** — parallel ridges <2 m high and metres apart, recording very rapid deformation beneath surging ice.

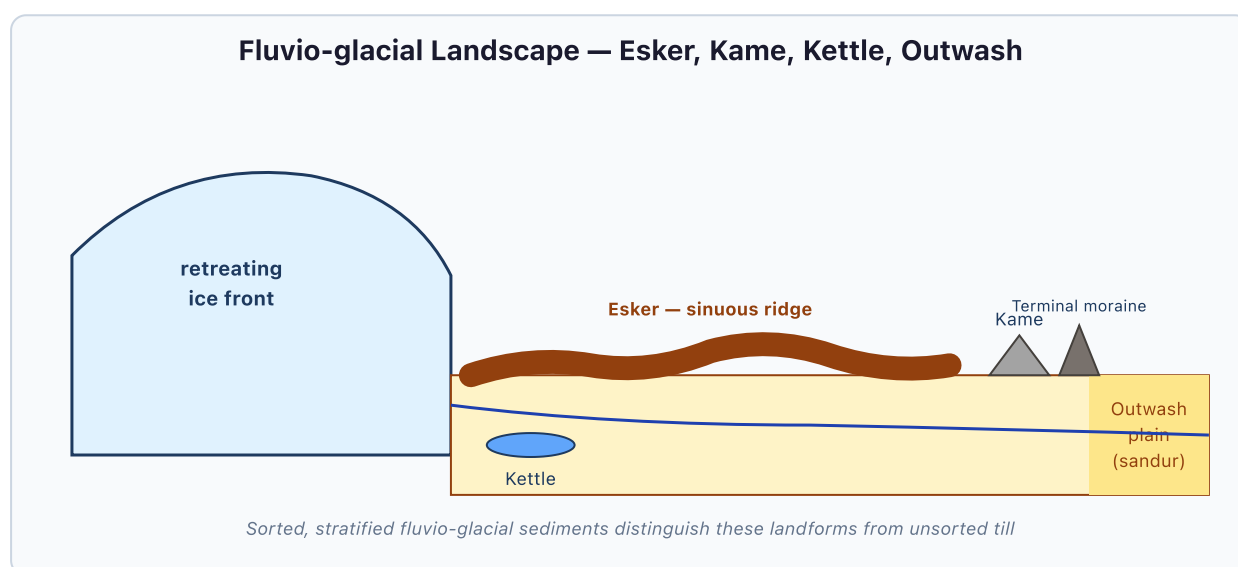
## 7. Fluvio-glacial Landforms

Meltwater — supraglacial, englacial, subglacial and pro-glacial — reworks till and produces sorted, stratified deposits. These **fluvio-glacial** landforms are diagnostic of wet-based glaciation and of the transition from full glaciation to deglaciation.

**Esker:** A long, sinuous ridge of sorted and stratified sand and gravel, formed by deposition inside a subglacial or englacial meltwater tunnel. When the surrounding ice melts, the tunnel-fill is let down onto the land surface as a raised ridge. Eskers ignore topography (they climb gentle slopes) because they were moulded by pressurised meltwater in enclosed channels.

- **Eskers** — sinuous ridges of stratified sand and gravel, up to tens of metres high and kilometres long. Deposited inside tunnels beneath or within the ice. Good example: Trim (Ireland); major Swedish and Finnish networks used as road alignments.
- **Kames** — irregular mounds of stratified sediment deposited by meltwater in contact with stagnant ice (e.g. in crevasses, supraglacial lakes, or along ice margins). *Kame terraces* form along valley sides where meltwater ran in a trench between the ice and the valley wall.
- **Kettle holes** — depressions formed when a block of stagnant ice, buried within fluvio-glacial sediment, eventually melts. Often water-filled — Thornton Loch (Scottish Borders), thousands in the Great Lakes region.

- **Outwash plains (sandur)** — extensive, gently sloping plains of sorted and stratified sand and gravel, built up by braided meltwater streams beyond the terminal moraine. Icelandic examples are the type-locality — Skeiðarársandur (~1,300 km<sup>2</sup>) was periodically remodelled by jökulhlaups from Grímsvötn.
- **Pro-glacial lakes** — meltwater lakes dammed at the ice margin, between the ice and either a moraine or higher topography. Late-Pleistocene Lake Agassiz in North America was larger than all modern Great Lakes combined; Lake Lapworth in the English Midlands carved the Ironbridge Gorge as it overspilled.



## Diagnostic Differences: Till vs Fluvio-glacial Sediment

| Property          | Till (direct ice deposit)            | Fluvio-glacial (meltwater deposit)             |
|-------------------|--------------------------------------|------------------------------------------------|
| Sorting           | Unsorted (clay to boulders together) | Well sorted by size along the flow             |
| Stratification    | None or rare                         | Prominent bedding planes, cross-stratification |
| Clast shape       | Angular to sub-angular               | Rounded to well-rounded                        |
| Clast orientation | Fabric often aligns with ice flow    | Imbrication along water flow                   |

|                |                                     |                                          |
|----------------|-------------------------------------|------------------------------------------|
| Matrix         | Mud/silt-dominated (lodgement till) | Sand or gravel, usually clean            |
| Landform hosts | Moraines, drumlins, ground moraine  | Eskers, kames, outwash, varved lake beds |

## 8. Periglacial Environments

**Periglacial** environments occupy the margins of glaciated regions, where ground is frozen but not ice-covered. They cover ~25% of Earth's land area today (high Arctic, alpine summits, parts of Antarctica not under ice). In Britain, periglacial processes dominated ice-free lowlands during Pleistocene cold stages and have left a diagnostic *relict periglacial* legacy on Dartmoor and the North York Moors. The defining characteristic is **permafrost** and an overlying seasonally thawed **active layer**.

**Periglacial:** A cold-climate environment characterised by permafrost, intense freeze-thaw activity and an absence of permanent ice cover. Processes include frost weathering, frost heave, solifluction, nivation and ground ice development. Does not strictly imply proximity to glaciers — many periglacial areas (e.g. Siberia) have no glaciers.

**Permafrost:** Ground (soil, sediment or rock) that remains at or below 0 °C continuously for at least two consecutive years. Classified by continuity: *continuous* (>90% of ground underlain, mean annual air temperature < −8 °C), *discontinuous* (50–90%), and *sporadic* (<50%). Can exceed 1,500 m thickness in Siberia.

## Periglacial Processes

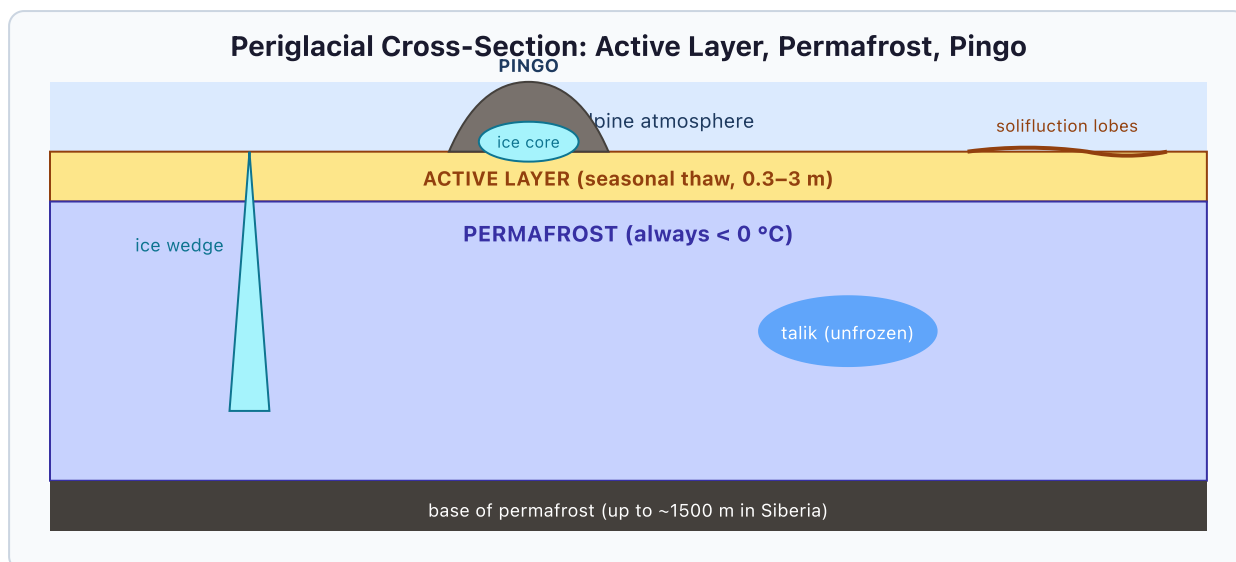
| Process                         | Description                                                                    | Associated landform                        |
|---------------------------------|--------------------------------------------------------------------------------|--------------------------------------------|
| Frost weathering (gelifraction) | Water freezes in rock joints; 9% volume expansion fractures rock               | Blockfields (felsenmeer), tors, scree      |
| Frost heave                     | Ice-lens growth pushes clasts vertically within soil; large stones move upward | Patterned ground, stone circles, polygons  |
| Solifluction (gelifluction)     | Slow downslope flow of saturated active layer over permafrost                  | Solifluction lobes/terraces, head deposits |

|                      |                                                              |                                                   |
|----------------------|--------------------------------------------------------------|---------------------------------------------------|
| Ground ice formation | Ice wedges, ice lenses, pingos from injection or segregation | Ice-wedge polygons, pingos, palsas                |
| Nivation             | Freeze-thaw beneath and beside snow patches                  | Nivation hollows (proto-cirques)                  |
| Thermokarst (thaw)   | Permafrost degrades, ground subsides unevenly                | Thaw lakes, slump scars, collapsed infrastructure |

## Periglacial Landforms

- **Pingos** — conical ice-cored hills, 30–70 m high. *Open-system* (Greenland) pingos form where artesian groundwater freezes beneath a cap; *closed-system* (Mackenzie Delta) pingos form where a talik beneath a drained lake progressively refreezes, expelling water to form a near-surface ice lens.
- **Patterned ground** — stone circles, polygons, stripes, nets produced by frost heave sorting and contraction cracking.
- **Ice-wedge polygons** — polygonal networks (4–40 m diameter) outlined by ice-filled contraction cracks that thermal stress re-opens each winter.
- **Blockfields (felsenmeer)** — near-horizontal surfaces covered in angular frost-shattered rubble. Cairngorm summits preserve classic relict examples.
- **Tors** — isolated bedrock outcrops on interfluvies, produced by differential weathering followed by periglacial stripping of the rotted debris. Dartmoor is the UK type-locality (Haytor, Hound Tor).
- **Solifluction deposits (head)** — sheets of mixed, poorly sorted sediment at the base of slopes.
- **Thermokarst** — an irregular, hummocky, lake-pitted landscape produced by permafrost thaw. Now spreading rapidly with climate warming.

**Pingo:** A conical, ice-cored hill of the periglacial landscape, formed where groundwater freezes beneath the surface as a lens of ice that pushes the overlying ground upward. “Open-system” pingos are fed by artesian water from surrounding slopes; “closed-system” pingos form in drained lake beds. Over 1,350 pingos occur in the Mackenzie Delta (NW Canada).



**Exam tip — permafrost is a climate archive:** Questions on periglacial environments often ask about climate change. Permafrost temperatures have risen  $\sim 0.3$  °C per decade since 1980. When permafrost thaws, soils release stored organic carbon as CO<sub>2</sub> and methane — a powerful positive feedback on warming. Always quote a quantified figure and name the feedback loop for top marks.

## Periglacial Processes Summary Table

| Landform                | Process                                      | Typical scale                  | UK / global example                                 |
|-------------------------|----------------------------------------------|--------------------------------|-----------------------------------------------------|
| Pingo                   | Ground-ice uplift<br>(open/closed system)    | 30–70 m high, 100–600 m across | Mackenzie Delta; relict pingos in Norfolk Breckland |
| Ice-wedge polygon       | Thermal contraction cracking refilled by ice | 4–40 m polygon width           | Siberian tundra; relict casts in East Anglia        |
| Patterned ground        | Frost heave sorting                          | cm to m scale                  | Cairngorm plateau (active); Dartmoor (relict)       |
| Blockfield (felsenmeer) | Frost weathering on low-relief summits       | plateau scale                  | Cairngorms, Prince of Wales Icefield                |
| Tor                     | Deep weathering then periglacial stripping   | 5–20 m                         | Haytor, Hound Tor (Dartmoor)                        |

|                   |                                              |                                |                                                |
|-------------------|----------------------------------------------|--------------------------------|------------------------------------------------|
| Solifluction lobe | Saturated active-layer creep over permafrost | 1–3 m high, 10s m wide         | Yukon plateau; slope heads in southern England |
| Thermokarst       | Permafrost thaw subsidence                   | km-scale hummocky lake pitting | Batagaika Crater; Yamal Peninsula              |

## 9. Relict Glaciated Landscapes of the UK

Britain has not been glaciated since the Younger Dryas ended ~11,700 years BP. Its glaciated landscapes are therefore *relict*: landforms and sediments inherited from the Pleistocene, now subject to Holocene fluvial and slope modification. Two areas dominate A-level syllabi.

### Snowdonia (Eryri)

- **Cwm Idwal** — a classic corrie with plucked back wall (idwal Slabs), over-deepened basin occupied by Llyn Idwal, and raised threshold. First NNR declared in Wales (1954).
- **Nant Ffrancon** — the archetypal U-shaped trough of the British syllabus: 6 km long, broad flat floor, steep walls, misfit Afon Ogwen, and the hanging valley of Cwm Idwal above it.
- **Llyn Padarn & Llyn Peris (Llanberis)** — ribbon lakes in a trough once occupied by a tongue of the Irish Sea ice stream and local ice.
- **Glyders & Crib Goch** — arêtes, pyramidal peak elements and frost-shattered summit blockfields.
- **Cwm Graianog** — well-preserved moraines and outwash mark the Younger Dryas (Loch Lomond Stadial) re-advance limits.

### Lake District (Cumbria)

- **Helvellyn massif** — the *Striding Edge* arête with the *Red Tarn* corrie beneath; its companion arête Swirral Edge defines a near-pyramidal peak.
- **Wastwater** — England's deepest lake (79 m), ribbon lake in a glacial trough, with extensive scree slopes showing periglacial activity.
- **Ullswater** — classic zigzagging ribbon lake occupying a trough with recessional moraines along its margins.
- **Langdale Pikes** — volcanic ridges shaped by ice, with hanging valleys and truncated spurs.



**Case Study 1 — Snowdonia National Park (Eryri):** Snowdonia covers 2,176 km<sup>2</sup> of North Wales and contains the most intensively studied relict glacial landscape in the UK. The central Snowdon massif preserves a complete suite of erosional landforms from the Dimlington (~22 ka) and Loch Lomond Stadial (~12.8–11.7 ka) glaciations: *cirques* (Cwm Idwal, Cwm Cau, Cwm Dyli), *arêtes* (Crib Goch, Y Lliwedd-Snowdon ridge), *pyramidal peak* (Snowdon/Yr Wyddfa itself is a multi-cirque-bounded pyramid), *U-shaped troughs* (Nant Ffrancon, Nant Peris, Dyffryn Mymbyr), *hanging valleys* (Cwm Idwal over Nant Ffrancon) and *ribbon lakes* (Llyn Padarn, Llyn Peris, Llyn Dinas). Terminal moraines of the Younger Dryas cirque glaciers are preserved in Cwm Graianog and Cwm Llydaw.

**Management:** designated a National Park in 1951; 5,500 km of public footpaths; ~4 million visitors annually (2022). Issues include footpath erosion on the Llanberis and Pyg tracks (scar-cutting visible from space), dry-stone wall and boundary maintenance, biodiversity (Snowdon lily, *Gagea serotina*, is a glacial relict), upland sheep farming under post-Brexit ELM schemes, and competing pressures of wilderness conservation vs development (Snowdon Mountain Railway, visitor centre Hafod Eryri at 1,085 m). Recent moves include voluntary car-free days in Ogwen Valley (2023) and adoption of Welsh-language place-name primacy (Eryri, Yr Wyddfa) from 2022.

## The Lake District as a Cultural Landscape

Since UNESCO inscription in 2017 the Lake District has been formally recognised as a cultural landscape shaped jointly by ice and by 1,000 years of pastoral farming. The glaciated landforms (16 ribbon lakes, >20 named arêtes, hundreds of corries) provide the stage; hefted Herdwick sheep, dry-stone walls, tarn-head barns and John Ruskin's aesthetic heritage provide the narrative. Management under the Lake District National Park Authority (est. 1951; 2,362 km<sup>2</sup>) must reconcile:

- **Biodiversity** — upland wetland protection; mountain ringlet butterfly (*Erebia epiphron*), the UK's only true montane butterfly, restricted to north-facing corrie slopes.
- **Carbon sequestration** — peatland restoration on the fells (Kinder Scout-style rewetting rolled out in Ullswater catchment 2022).
- **Flood mitigation** — natural flood management in tributary corrie-head valleys feeding Storm Desmond (December 2015) lowlands downstream (341.4 mm rainfall in 24 h at Honister, UK record).
- **Affordable housing** — second-home ownership in Cumbria reaches 20% in some parishes, pricing locals out (Levelling-up and Regeneration Act 2023 includes new planning powers).

## 10. Active Glaciated Landscapes

Unlike the UK, several high- and mid-latitude regions preserve *active* glaciers. Eduqas requires candidates to contrast these with relict landscapes and discuss the risks they generate.

| Region                 | Glaciers                                                          | Hazards / features                                  |
|------------------------|-------------------------------------------------------------------|-----------------------------------------------------|
| European Alps          | ~4,000; all temperate; fastest retreat in recorded history        | Rock-ice avalanches; ski-industry disruption; GLOFs |
| Himalayas & Karakoram  | ~54,000 glaciers; "Third Pole"; 1.4 billion downstream            | GLOFs (Chorabari 2013); water-security risk         |
| Iceland                | Ice caps (Vatnajökull, Langjökull); subglacial volcanism          | Jökulhlaups; mass tourism                           |
| Patagonia              | Northern & Southern Icefields; fastest retreat 1985–2011 globally | Calving; sea-level contribution                     |
| Alaska / Coast Ranges  | Numerous valley and tidewater glaciers                            | Iceberg discharge; tourism; infrastructure thaw     |
| Antarctica & Greenland | Continental ice sheets; polar & polythermal                       | Dominant contributors to global sea-level rise      |

**Case Study 2 — The European Alps:** The Alps host ~4,000 glaciers covering ~1,800 km<sup>2</sup> in 2022, down from ~4,500 km<sup>2</sup> in 1850. The Swiss Rhone Glacier, once filling the valley above Gletsch, has retreated ~1.3 km since 1850 and ~350 m in the past decade alone; it now loses ~40 cm of thickness per summer. The Aletsch Glacier (largest in the Alps, 23 km long, 81 km<sup>2</sup>) is retreating ~50 m yr<sup>-1</sup>. *Konkordiaplatz*, the ice-flow confluence above Aletschgletscher, thinned 200 m between 1860 and 2020, exposing new bedrock in the classroom of every Alpine physical geographer.

**Hazard: glacial-lake outburst floods (GLOFs).** Retreating glaciers expose pro-glacial lakes dammed by unstable moraines; a single collapse can release 10<sup>6</sup>–10<sup>7</sup> m<sup>3</sup> of water (a related mountain hazard, the 2017 Piz Cengalo rock avalanche and debris flow at Bondo, Switzerland,

killed 8 people). Lac des Faverges and >100 new lakes have formed since 1970.

**Ski-industry adaptation:** Snow-cannon coverage has expanded to >50% of European pistes; upper pistes are relocated above ~2,400 m; protective white geotextiles cover ski-run glacier tongues over summer (e.g. Kitzsteinhorn, Austria); some resorts now operate summer skiing via tunnels bored beneath glacier tongues (Pitztal, Austria).

**Conservation tensions:** Expansion of the Rettenbachferner ski area clashed with Tyrolean environmental law in 2022; the Swiss Federal Council rejected all new cableway concessions on ice-covered terrain in 2024. Glacier Initiative (2023) commits Switzerland to net-zero by 2050 explicitly to defend remaining ice mass.

## Further UK Sites

- **Cairngorms** — relict arctic-alpine plateau; blockfields, patterned ground, solifluction lobes; Britain's highest continuous periglacial surface (Ben Macdui 1,309 m).
- **Glen Coe** — pyroclastic caldera deeply over-deepened; the Three Sisters and the Lost Valley are textbook truncated spurs and hanging valleys.
- **Dartmoor** — granite tors on a periglacial plateau: Haytor, Hound Tor, Vixen Tor; clitter slopes of frost-shattered granite rubble; blockfields on summits.
- **Yorkshire Dales** — Ribblesdale drumlin field (hundreds of drumlins in a 20 km stretch); Ingleborough limestone pavement above Norber erratics.
- **East Anglia** — terminal (push) moraine of the Anglian glaciation forms the Cromer Ridge; ice-wedge casts and pingo scars in the Brecklands.

## Peak Water

As glaciers shrink, their summer meltwater output initially rises (more exposed ice, more melt per unit surface) before declining to near zero once the reservoir is exhausted. The peak output — *peak water* — is a critical planning concept for glacier-fed hydrological systems. Most Alpine catchments are projected to pass peak water between 2030 and 2060; in the high Andes, peak water has already passed in many tropical glacier basins. Communities and infrastructure downstream must plan for a progressive decline in dry-season flow from the 2060s onward.

## 11. Climate Change and Glaciated Environments

Glaciers and permafrost are among the most sensitive indicators of anthropogenic warming. Key data:

- **Global glacier mass balance** has been persistently negative since 1980; total global glacier mass loss 2000–2019 was  $\sim 267 \pm 16 \text{ Gt yr}^{-1}$  (Hugonnet et al., 2021), contributing  $\sim 21\%$  of observed sea-level rise.
- **Greenland Ice Sheet** loses  $\sim 270 \text{ Gt yr}^{-1}$ ; Antarctic Ice Sheet  $\sim 150 \text{ Gt yr}^{-1}$ ; together they account for more of observed sea-level rise than thermal expansion.
- **Thwaites Glacier** (West Antarctica), dubbed the “Doomsday Glacier”, drains  $\sim 4\%$  of global sea-level rise currently; the Marine Ice Sheet Instability (MISI) hypothesis suggests its grounding-line retreat could be irreversible, contributing up to 65 cm to long-term sea level.
- **Permafrost** temperatures have risen  $\sim 0.3 \text{ }^{\circ}\text{C}$  per decade since 1980 (IPCC AR6, 2021). Projected thaw of shallow permafrost by 2100 is  $\sim 25\%$  under RCP2.6 and  $\sim 70\%$  under RCP8.5.
- **Alpine glacier retreat** — Rhone Glacier  $\sim 1.3 \text{ km}$  since 1850; Mer de Glace  $\sim 1.9 \text{ km}$  since 1850; a third of Alpine ice has gone since 2000.

### Marine Ice Sheet Instability (MISI)

Much of West Antarctica rests on a bed that slopes *downward inland* from its margin — known as a *retrograde slope*. Once the grounding line (the point where ice leaves the bed and begins to float) retreats past a threshold, deeper water produces thicker ice, thicker ice produces higher calving flux, calving flux accelerates retreat, and the glacier is locked into runaway collapse. Thwaites Glacier (drainage basin  $192,000 \text{ km}^2$ ) is considered the highest-risk system under MISI; if the West Antarctic Ice Sheet fully collapses through this mechanism, global sea level would rise  $\sim 3.3 \text{ m}$  over centuries. The Marine Ice Cliff Instability (MICI) hypothesis adds a second positive feedback, whereby tall unbuttressed ice cliffs fail by brittle fracture once they exceed  $\sim 90 \text{ m}$  subaerial height. Neither feedback is fully captured in current IPCC model ensembles, leaving high-end sea-level projections uncertain.

### Climate Change and UK Glacial Features

Although no glaciers remain in Britain, Holocene climate change continues to modify the relict landscape. Scree slopes in Wasdale are now largely inactive; peat cover on Snowdonia's blockfields has thickened; tarns accumulate sediment at  $\sim 1 \text{ mm yr}^{-1}$ . Projected warming of  $2\text{--}4 \text{ }^{\circ}\text{C}$  by 2100 threatens

arctic-alpine biota: *Saxifraga cespitosa* on Ben Lawers and the Snowdon lily (*Gagea serotina*) are now the subject of conservation translocation schemes.

## Consequences and Feedbacks

- **Sea-level rise** — AR6 Central estimate 0.44–0.76 m by 2100 (SSP2-4.5).
- **Glacial-lake outburst floods (GLOFs)** — frequency tripled since 1970s.
- **Water-security** — meltwater-fed rivers (Indus, Ganges, Amu Darya, Po, Rhine) face summer flow reduction past 2050.
- **Permafrost carbon feedback** — ~1,500 Gt of carbon stored; thaw releases CO<sub>2</sub> and CH<sub>4</sub>, amplifying warming.
- **Thermokarst** — ground subsidence damages pipelines, roads and buildings across the Arctic.
- **Albedo feedback** — loss of white ice cover lowers regional albedo, accelerating warming.

### Case Study 3 — Arctic Permafrost: Siberia and Alaska:

**Siberia:** >60% of Russia underlain by permafrost. 2020 heatwave (Verkhoyansk reached 38 °C, Arctic record); thaw subsidence damaged the Norilsk diesel tank (29 May 2020), releasing 21,000 tonnes of fuel into the Ambarnaya River — Russia's worst Arctic oil spill, directly attributed to foundation failure of an ageing storage tank as permafrost degraded. Thaw lakes have expanded >15% in some West Siberian regions since 1990. Yedoma (Pleistocene carbon-rich loess) thaw releases methane by microbial decay; the Batagaika “megaslump” in Yakutia has grown to 1 km long, 100 m deep, expanding at 10–30 m yr<sup>-1</sup>.

**Alaska:** ~80% underlain by permafrost. Alaska Highway and Dalton Highway show chronic “drunken forest” subsidence; repair costs estimated at US\$4.6 billion by 2100 (ACIA, 2019). Trans-Alaska Pipeline is elevated on >124,000 heat-pipe-cooled piles specifically to avoid permafrost thaw beneath.

**Indigenous community case — Shishmaref:** Iñupiat village of ~600 on the Chukchi Sea coast; sea-ice loss has exposed the coastline to storm waves previously absorbed by winter fast ice; combined with permafrost thaw, the village has lost >30 m of coast in the past 30 years. 2016 vote (89 to 78) to relocate inland; estimated cost US\$180 million; still unrealised in 2026. Shishmaref has become a global reference case for “climate displacement” under international climate-justice

law. The Biden-Harris 2022 *Community-Driven Relocation* fund ring-fenced US\$25 million for Alaska Native coastal villages.

**Case Study 4 — Iceland (Vatnajökull):** Iceland's glaciers cover ~10% of the island; Vatnajökull is Europe's largest ice cap by volume (~3,100 km<sup>3</sup>, 7,900 km<sup>2</sup>) and sits atop the Grímsvötn and Bárðarbunga volcanic systems — a textbook coupling of glacial and tectonic geomorphology. Volume loss has averaged ~0.8% yr<sup>-1</sup> since 1995; Okjökull was declared “dead” (no longer a glacier) in 2014, formally memorialised with a plaque in 2019.

**Jökulhlaups (subglacial outburst floods):** recurrent at Skeiðarársandur where the Grímsvötn subglacial caldera melts ice, fills with meltwater, then suddenly drains beneath the ice cap. The November 1996 jökulhlaup following the Gjálp sub-ice eruption discharged ~45,000 m<sup>3</sup>s<sup>-1</sup> at peak (roughly a fifth of the mean flow of the Amazon), destroyed 10 km of the Ring Road (Route 1), two bridges, and several electricity pylons. Reconstruction cost ~ISK 850 million. Modern protection: relocated bridges on raised piers; real-time subglacial GPS, seismic and lake-level monitoring by the Icelandic Meteorological Office.

**Glacier tourism:** Vatnajökull National Park (est. 2008, UNESCO World Heritage 2019) receives ~700,000 visitors yr<sup>-1</sup>. Jökulsárlón glacial lake has grown from ~8 km<sup>2</sup> (1975) to >18 km<sup>2</sup> (2022); glacier-guide trips operate on Solheimajökull (ice-cave tours) but face shifting safety constraints as the snout retreats.

**Geothermal-glacial interactions:** sub-ice melt over the Grímsvötn caldera pre-loads meltwater that triggers eruption pressure-release cycles; the Eyjafjallajökull eruption (2010) and its ash plume that closed much of European airspace for about 6 days was initiated by a sub-glacial eruption that produced substantial phreatomagmatic fragmentation via ice-melt interaction.

**Exam tip — quantified climate data earn AO1 marks:** Recent examiner reports explicitly reward candidates who deploy named quantitative data: glacier mass loss (267 Gt yr<sup>-1</sup>), permafrost warming (0.3 °C / decade), Rhone Glacier retreat (1.3 km since 1850), Aletsch thinning. “Glaciers are melting faster now” is an AO1 statement worth one mark; “Hugonnet et al. (2021) measured global glacier loss at 267 Gt yr<sup>-1</sup> contributing 21% of sea-level rise” is worth three.



## 12. Management of Active and Relict Glaciated Environments

Management challenges divide into three classes: (i) *use* of mountain landscapes for tourism, recreation and resource extraction; (ii) *hazard* management (GLOFs, ice avalanches, jökulhlaups, thaw subsidence); and (iii) *conservation* of fragile environments under climate change.

### Case Studies Summary Matrix

| Theme                 | Snowdonia                | Alps                       | Siberia / Alaska               | Iceland                     |
|-----------------------|--------------------------|----------------------------|--------------------------------|-----------------------------|
| Ice regime            | Relict (gone ~11.7 ka)   | Active temperate           | Periglacial (permafrost)       | Active sub-polar ice caps   |
| Key landform          | Nant Ffrancon trough     | Aletsch Glacier            | Yedoma / ice-wedge polygons    | Vatnajökull, sandur         |
| Dominant hazard       | Footpath erosion         | GLOFs, rock-ice avalanches | Thaw subsidence, methane       | Jökulhlaups, eruptions      |
| Key management        | NP 1951, ELM schemes     | Monitoring, snowmaking     | Elevated pipelines, relocation | Real-time monitoring        |
| Climate-change signal | Arctic-alpine biota loss | 1.3 km Rhone retreat       | Norilsk 2020 fuel spill        | Okjökull declared dead 2014 |

### Tourism vs Conservation

Both relict (Snowdonia, Lake District, Cairngorms) and active (Alps, Iceland, Alaska) glaciated landscapes draw mass tourism. Benefits: rural employment, cultural visibility of protected areas, visitor-fee funding for conservation. Costs: footpath erosion, increased carbon footprint of access travel, infrastructure sprawl, cultural dilution of host communities.

The **Cairngorms National Park** (Scotland) protects 4,528 km<sup>2</sup> of arctic-alpine plateau with relict periglacial features (blockfields, patterned ground, frost boils), 55 Munros and Britain's only natural *Calluna-Cladonia* cloud-forest analogue. Issues: the Cairngorm Funicular controversy (closed 2018, Highland Council £16 million repair 2023); proposed deer-cull management under Cairngorms 2045 strategy; balance between rewilding (beaver reintroduction, 2022) and active estate management.

## Carrying Capacity and Visitor Management

Every popular glaciated landscape has a *physical carrying capacity* (what the landform can absorb without degradation) and a *perceptual carrying capacity* (at what visitor density the experience of wilderness is lost). The Snowdon summit hosts ~600,000 ascents a year and visible path erosion has required >£1 million resurfacing since 2000. Management techniques include honey-pot concentration (Hafod Eryri visitor centre), zoning (quiet zones in Ogwen), timed-entry booking (Mount St Helens, Cadair Idris trial 2024) and voluntary "responsible-access" codes (Scottish Outdoor Access Code 2005). Overtourism is now framed in the policy literature as a *glacial-specific* issue because loss of glaciers intensifies the scarcity value of those that remain.

## Hydropower

Glaciated valleys offer high gradients, narrow sites for dams, and steady meltwater flow, making them historically attractive for hydropower. Norway (~88% of electricity), Iceland (~70%, remainder geothermal) and Switzerland (~60%) derive electricity from glacier-fed schemes. Trade-offs: flooded trough landscapes (Dinorwig, Wales; Glen Affric, Scotland), fish passage interruption, sediment-trap effects below reservoirs. Climate change brings a medium-term flow boost as glaciers melt, followed by a long-term flow collapse when ice reserves are exhausted — the so-called "*peak water*" trajectory (typically 2030–2060 for Alpine basins).

## Indigenous and Cultural Rights

High-latitude glaciated and periglacial environments are home to indigenous peoples whose land rights and livelihoods are directly threatened: Inuit in Greenland and Canada, Iñupiat and Yup'ik in Alaska, Sami in Fennoscandia, Evenki and Nenets in Siberia. Climate adaptation and mitigation agreements (UNDRIP 2007, Paris 2015) now require meaningful consultation; in practice, forced relocations (Shishmaref, Newtok) demonstrate the gap between declared rights and delivery.

## Mineral and Hydrocarbon Extraction

Periglacial environments hold significant mineral wealth: nickel and palladium at Norilsk (Siberia), gold at Kumtor (Kyrgyzstan, on the Davidov Glacier), and uranium at Kiggavik (Nunavut, Canada). Extraction is economically powerful but environmentally destructive and carbon-intensive: the Kumtor mine has dumped rock waste directly onto active glacier ice since 1995, covering >20 km<sup>2</sup> and accelerating ice melt. Oil and gas pipelines cross permafrost by thousands of kilometres: the Alyeska Pipeline (1,287 km) uses elevated heat-pipe piling; the Power of Siberia natural gas pipeline to China (3,000 km) is re-



engineering permafrost management practice in the 2020s. Decarbonisation policies (Paris 2015, UK Climate Change Act) now challenge the long-term economics of these projects.

## Ski-area Development

Controversial tension between local economic dependence on winter sports and environmental damage from cableway pylons, piste engineering and artificial snow. In the Alps, new glacier-ski developments require environmental impact assessment under EU law; proposed tunnels under glacier tongues (e.g. Pitztal-Ötztal connection) have been contested by ProNatura Switzerland. In Scotland, Cairngorm Mountain's ski operation depends increasingly on artificial snow, raising questions about viability below 900 m by 2050.

**Exam tip — structure evaluation questions with three pillars:** For 20-mark “to what extent” questions on management, consistently evaluate along environmental, economic and social dimensions, then integrate a climate-change time-axis (short-term, medium-term, long-term). This structure is directly rewarded under AO3 and allows examiners to follow your argument.

**Exam tip — evaluate sustainability explicitly:** In 20-mark essays on management, examiners reward a final conclusion that names the *most* and *least* sustainable option with justification. For example: “Of the three options — hard engineered protection, nature-based GLOF drainage, and managed retreat — managed retreat is the most environmentally sustainable but the least socially acceptable in the short term; medium-term nature-based management offers the best balance.”

## Hazard Management

Active glaciated and periglacial landscapes pose a characteristic suite of hazards that management regimes must anticipate:

- **Glacial-lake outburst floods (GLOFs)** — managed through monitoring, controlled siphoning (Tsho Rolpa, Nepal), and engineered moraine breach (Imja, Nepal, 2016).
- **Rock-ice avalanches** — triggered by permafrost thaw in steep rock walls; fatal events include Kolka-Karmadon (Russia 2002, 125 dead) and Val Pola (Italy 1987, 27 dead). Management includes retreat from affected zones and early-warning seismic networks.
- **Jökulhlaups** — monitored by the Icelandic Meteorological Office and Norwegian NVE using subglacial geophysics, seismicity and GPS.

- **Slope instability from permafrost thaw** — Alpine huts, cable stations and pipelines increasingly require structural stabilisation as the permafrost beneath their foundations warms (e.g. Swiss “Matterhorn rockslide 2003” closed ascent for a season).
- **Infrastructure damage in periglacial settings** — thermosyphons, insulated piling, elevated construction (trans-Alaska Pipeline) and seasonal road closures (Siberian *zimniki*).

## 13. Contested Management — Wilderness vs Development

A persistent theme in Eduqas case studies is the contest between *wilderness* value (intrinsic, aesthetic, scientific) and *development* pressure (economic, recreational, extractive). Key framings:

- **Wilderness as resource** — Snowdonia Dark Sky Reserve (2015), IUCN Category II protection, biodiversity reservoirs.
- **Wilderness as social construct** — critics (William Cronon, 1996) argue “wilderness” erases indigenous histories; in the Alps, centuries of pastoral transhumance contradict any “pristine” baseline.
- **Upland farming** — the Herdwick sheep tradition in the Lake District, hefted to fells, is protected under the UNESCO Cultural Landscape designation (2017). Post-Brexit ELM schemes pay farmers for “public goods” (landscape, biodiversity, carbon).
- **Protected-area networks** — 15% of Alpine area is designated (Natura 2000, IUCN categories); still crossed by 20,000 km of cableway.
- **Climate-change adaptation** — managed retreat of ski resorts, voluntary carbon offsetting (Dolomite 2026 Olympic pledge), relocation of mountain shelters (Alpine huts built on moving rock now face foundation instability as permafrost thaws).

### Stakeholder Analysis Framework

| Stakeholder                  | Priority                           | Typical conflict                          |
|------------------------------|------------------------------------|-------------------------------------------|
| National park authority      | Conservation, visitor management   | vs ski operators wanting expansion        |
| Hill farmers (e.g. Herdwick) | Grazing rights, subsidy continuity | vs rewilders supporting tree regeneration |

|                                      |                                             |                                                   |
|--------------------------------------|---------------------------------------------|---------------------------------------------------|
| Tourism operators                    | Access, infrastructure                      | vs visitor-carrying-capacity limits               |
| Indigenous communities (Sami, Inuit) | Land rights, cultural continuity            | vs resource extraction (oil, mines)               |
| Energy companies                     | Hydropower, mineral access                  | vs wilderness protection                          |
| Climate scientists                   | Research access, monitoring infrastructure  | vs park "naturalness" restrictions on field sites |
| Local residents                      | Employment, services                        | vs second-homeowners driving up housing costs     |
| National government                  | International climate commitments, heritage | vs short-term economic pressures                  |

Analysis of stakeholder conflicts is rewarded under AO2 ("apply geographical knowledge and understanding to interpret and evaluate") and AO3 ("use a variety of relevant quantitative, qualitative and fieldwork skills to investigate questions and issues, reach evidenced conclusions"). Always name at least three stakeholders with opposing priorities and state your own judgement at the end of a 20-mark answer.

## 14. Synoptic Summary

Glaciated landscapes (Eduqas 1.2) are dynamic systems whose past operation has shaped much of temperate Britain, and whose present operation at high latitudes and altitudes is a leading indicator of anthropogenic climate change. Top-band answers will treat a glacier as an open system in dynamic equilibrium, link *process* (plucking, abrasion, meltwater deposition) explicitly to *landform* (cirque, U-shaped trough, esker), and evaluate management through environmental, economic, social and long-term climate-change dimensions. Quantified case studies (Snowdonia, the Alps, Arctic permafrost in Siberia/Alaska, Vatnajökull) provide the evidence base.

| Case study | Type                   | Key facts to memorise                                                              |
|------------|------------------------|------------------------------------------------------------------------------------|
| Snowdonia  | Relict, temperate lat. | 2,176 km <sup>2</sup> ; Cwm Idwal; Nant Ffrancon; Crib Goch; ~4M visitors; NP 1951 |

|                       |                    |                                                                                                            |
|-----------------------|--------------------|------------------------------------------------------------------------------------------------------------|
| European Alps         | Active, temperate  | ~4,000 glaciers; Rhone –1.3 km since 1850; Aletsch –50 m yr <sup>-1</sup> ; Piz Cengalo 2017               |
| Siberia/Alaska        | Active periglacial | Norilsk 2020 spill; Shishmaref relocation; Alaska Highway thaw; 1,500 Gt permafrost C                      |
| Iceland (Vatnajökull) | Active subpolar    | 7,900 km <sup>2</sup> ; 1996 jökulhlaup 45,000 m <sup>3</sup> s <sup>-1</sup> ; Okjökull 2014; UNESCO 2019 |

## Command-Word Quick-Reference

| Command word              | What is being tested          | Suggested structure                                                                            |
|---------------------------|-------------------------------|------------------------------------------------------------------------------------------------|
| Define (2 marks)          | AO1 recall                    | One-sentence definition with a key quantitative criterion                                      |
| Explain (6–8 marks)       | AO1 + AO2 process chains      | Stage → process → cause → resulting landform                                                   |
| Assess (12 marks)         | AO2 + AO3 balanced evaluation | Intro judgement → argue for / argue against → conclusion                                       |
| To what extent (20 marks) | AO1 + AO2 + AO3 full essay    | Define terms → systems framework → 3 paragraphs with case studies → climate change → judgement |
| Compare / contrast        | AO2 application               | Same feature in two named places with quantitative difference                                  |

### Key revision checklist:

- Can you define and apply *mass balance*, *ELA*, *dynamic equilibrium*, *thermal regime*?
- Can you explain basal sliding, internal deformation, extending/compressive flow?
- Can you draw labelled diagrams for cirque formation, U-shaped valley cross-section, drumlin swarm, pingo cross-section?
- Can you distinguish till from fluvio-glacial sediment and give a landform example of each?
- Can you name four relict UK glaciated landforms with specific Snowdonia or Lake District locations?
- Can you quote 3–4 quantified facts per case study (Snowdonia, Alps, Arctic permafrost, Iceland)?

- Can you evaluate a glaciated-landscape management issue on environmental, economic and social sustainability?
- Can you integrate IPCC AR6 data on glacier mass loss and sea-level contribution into a climate-change paragraph?

**Exam tip — use active verbs for processes:** In process descriptions, use active verbs:

"meltwater *percolates* into joints and *refreezes*, *bonding* rock to ice; subsequent advance *extracts* the rock" is better than "plucking happens when meltwater goes in joints." Examiner reports specifically cite vague process descriptions as the single commonest cause of mark loss in process questions.

**Exam tip — always contrast relict with active:** Eduqas questions almost invariably reward candidates who compare a relict UK landscape (Snowdonia or Lake District) with an active high-latitude or high-altitude example (Alps, Iceland, Patagonia). Naming four diagnostic landforms from each, with quantified data, reliably hits AO1 and AO2 targets.